MILITARY SPECIFICATION SHEET

CHAFF ROLL (TYPE RR-171/ALE)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the chaff package described herein shall consist of this document and the latest issue of Specification MIL-C-85001.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, manufacture, and acceptance requirements for a chaff countermeasures designated RR-171/ALE. This equipment consists of a chaff roll set comprised of two chaff rolls loaded with aluminum coated glass chaff capable of producing confusing signals in radar systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-85001	Chaff Roll (Type RR-( )/ALE), General Specification for
MIL-R-60346	Roving, Glass, Fibrous

2.1.1 Availability of Documents. When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, CODE 105, 5801 Tabor Avenue, Philadelphia, PA 19120.

3. REQUIREMENTS

3.1 General Requirements. The RR-171/ALE chaff roll set shall comply with the requirements of MIL-C-85001 and this specification.

3.1.1 Item Definition. The chaff roll set covered by this specification shall consist of two chaff roll assemblies loaded with aluminum coated glass dipoles of the quantity and type described herein. This chaff roll set shall be functionally compatible with the AN/ALE-41 Dispensing Set.

3.2 Performance Characteristics. Performance characteristics shall comply with the requirements of MIL-C-85001.

FSC 5865

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3.3 Detail Requirements.

3.3.1 Chaff Roll Set. The RR-171/ALE chaff roll set shall consist of two identifiable chaff rolls consisting of the following:

3.3.1.1 Roll Assembly. Each chaff roll shall be contained on the roll assembly detailed in MIL-C-85001.

3.3.1.2 Chaff Roll Film. The chaff elements shall be placed between two layers of polyethylene terephthalate film (detailed in MIL-C-85001) to form a film/chaff/film sandwich.

3.3.1.3 Chaff Dipoles. The lengths and counts of chaff dipoles included within each roll of the chaff roll set shall be as listed in Table I. Linear interpolation is permissable.

DIPOLE LENGTH (inches)	TOTAL NUMBER IN ROLL
RR-171/ALE ROLL SET, ROLL 1	
1.99 ± .02	41 X 10^6 ± 10%
0.61 ± .02	164 X 10^6 ± 10%
0.71 ± .02	164 X 10^6 ± 10%
0.82 ± .02	123 X 10^6 ± 10%
1.12 ± .02	41 X 10^6 ± 10%
RR-171/ALE ROLL SET, ROLL 2	
1.74 ± .02	39 X 10^6 ± 10%
0.36 ± .02	468 X 10^6 ± 10%
0.45 ± .02	273 X 10^6 ± 10%
0.82 ± .02	39 X 10^6 ± 10%
1.12 ± .02	39 X 10^6 ± 10%

TABLE I. CHAFF DIPOLE LENGTHS AND COUNTS

3.3.1.3.1 Dipole Material. The dipole material shall be glass monofilament coated continuously with aluminum.

3.3.1.3.1.1 Glass Monofilament. The glass monofilament shall be drawn from Type E glass marbles made from a composition similar to that delineated in MIL-R-60346. The diameter of the glass monofilament shall be no larger than 0.95 mil.

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3.3.1.3.1.2 Aluminum Coating. The aluminum coating material shall be 99% pure aluminum. The coating thickness shall be a minimum of 0.06 mil., and shall be continuous along the length and around the circumference of the glass monofilament dipole. The weight of the coated monofilament shall be 390 ± 40 grams per foot of chaff cut (750,000 dipoles average count).

3.3.1.4 Coating. A slip coating shall be applied to the dipole material in accordance with the requirements of MIL-C-85001.

3.4 Assembly.

3.4.1 Film/Chaff/Film Sandwich. Each film/chaff/film sandwich shall be fabricated by laying the various chaff cuts in rows along the polyethylene film as illustrated in Figure 1. The rows shall be parallel to the leading edge of the chaff roll, and the individual chaff elements shall be perpendicular to the leading edge of the chaff roll. The 1.99 inch dipoles of roll 1, and the 1.74 inch dipoles of roll 2 shall be placed in the row nearest the leading edge of the film, so that when the film/chaff/film sandwich is wound onto the hub, these dipoles shall be at the leading edge of their respective rolls.

3.4.2 Chaff Roll. The chaff roll shall be fabricated in accordance with the requirements of MIL-C-85001.

3.4.3 Chaff Roll Weight. The minimum gross weight of the film, hub, and chaff combination shall be 43 pounds for Roll 1 and 41 pounds for Roll 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Inspection and testing of the RR-171/ALE chaff roll (set) shall be conducted in accordance with the requirements of MIL-C-85001.

5. PREPARATION FOR DELIVERY

5.1 General. The chaff roll (set) shall be prepared for delivery in accordance with the requirements of MIL-C-85001.

5.1.1 Item Description. The item description shall be "CHAFF SET (ROLL 1), RR-171/ALE" or "CHAFF SET (ROLL 2), RR-171/ALE" as applicable.

5.1.2 Marking. Each inner carton and barrier bag shall be marked with the item description and the words DO NOT SEPARATE in 1 inch red letters as detailed in Drawing 1050AS111.

6. NOTES

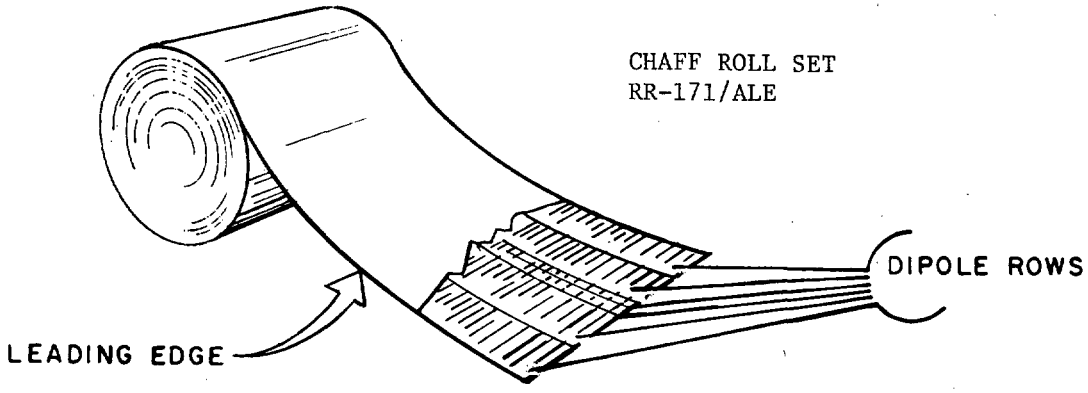
6.1 Notes. The notes contained in MIL-C-85001 shall apply.

Preparing Activity

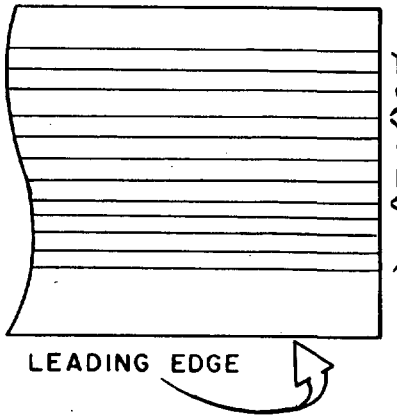
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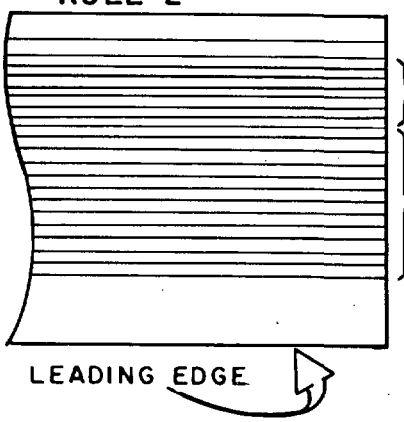


ROLL 1



DIPOLE LENGTH (INCHES)	NUMBER OF ROWS
1.12	1
0.82	3
0.71	4
0.61	4
1.99	1

ROLL 2



DIPOLE LENGTH (INCHES)	NUMBER OF ROWS
1.12	1
0.82	1
0.45	7
0.36	12
1.74	1

FIGURE 1

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-85001/171(AS) CHAFF ROLL (TYPE RR-171/ALE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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